DETECTING ANOMALOUS ACTIVITY OF A SHIP ENGINE

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Problem Overview

A shipping supply company aims to detect anomalous engine functionality, which represents unusual behavior that may affect ship engine performance. The dataset contains six engine-related features, each representing an indicator of engine condition and performance.

An anomaly is defined as an unusual combination of feature values across these engine features. Such anomalies may pose significant risks, including increased fuel consumption, reduced efficiency, delays in deliveries, and negative impact on revenue and customer satisfaction.

To address this problem, this project aims to develop an anomaly detection system capable of identifying potential engine problems at an early stage. Detecting these issues in advance allows the company to take preventive actions before serious damage occurs.

The approach combines exploratory data analysis, statistical techniques, and machine learning models to identify abnormal patterns in engine behavior. By applying these methods, the company can detect malfunctioning ship engines, reduce maintenance costs, minimize downtime, and improve operational safety.

Approach

The analysis started with exploratory data analysis to understand the distribution of the six engine performance features. A statistical summary table was generated showing the count, mean, standard deviation, minimum, maximum, and quartiles for each feature [Table 1 - Statistical Summary]. This table helped identify the normal operating ranges of engine rpm, oil pressure, fuel pressure, coolant pressure, oil temperature, and coolant temperature.

From the statistical summary [Table 1 - Statistical Summary], it was observed that most values lie within a consistent range, while some maximum values are much higher than the upper quartile, suggesting the presence of possible outliers. These unusual values indicate that some ships may operate under abnormal engine conditions, which makes anomaly detection necessary.

After the initial statistical analysis, the data was scaled to ensure all features contributed equally to the models. Principal Component Analysis (PCA) was then applied to reduce the dataset to two dimensions for visualization purposes. This allowed the main cluster of normal observations and potential anomalies to be clearly seen.

Two anomaly detection methods were applied: One-Class SVM and Isolation Forest. Different parameters were tested, and the best results were obtained using Isolation Forest with contamination = 0.01 and 100 estimators, which produced an anomaly rate within the required range. The detected anomalies were plotted on the PCA projection to confirm that abnormal points appear outside the main cluster.

Histograms and boxplots [Figure 1 - Boxplot and Distribution of Features] were also used to study the distribution of engine features. Most values lie within a normal range, forming dense clusters, while several outliers appear in the boxplots, especially for engine rpm and oil pressure. These extreme values indicate possible abnormal engine behavior and justify the use of anomaly detection methods.

Table 1 - Statistical Summary

index	count	mean	std	min	25%	50%	75%	max
Engine rpm	19535.0	791.2392628615306	267.6111931048414	61.0	593.0	746.0	934.0	2239.0
Lub oil pressure	19535.0	3.3037751547801384	1.0216429993848934	0.003384113	2.5188147159999996	3.1620347	4.0552715195	7.265565536
Fuel pressure	19535.0	6.6556154508940875	2.7610206475560792	0.003187131	4.916886097	6.20171956	7.7449733335	21.13832551
Coolant pressure	19535.0	2.3353688916173536	1.0363821029008646	0.002482733	1.6004658630000002	2.166882939	2.848839661	7.478504946
lub oil temp	19535.0	77.6434200131272	3.110983937271963	71.32197369	75.72599023000001	76.81735037	78.071690565	89.58079551
Coolant temp	19535.0	78.42743301515331	6.206749286559426	61.67332472	73.89542093	78.34666232	82.915411335	195.5279116

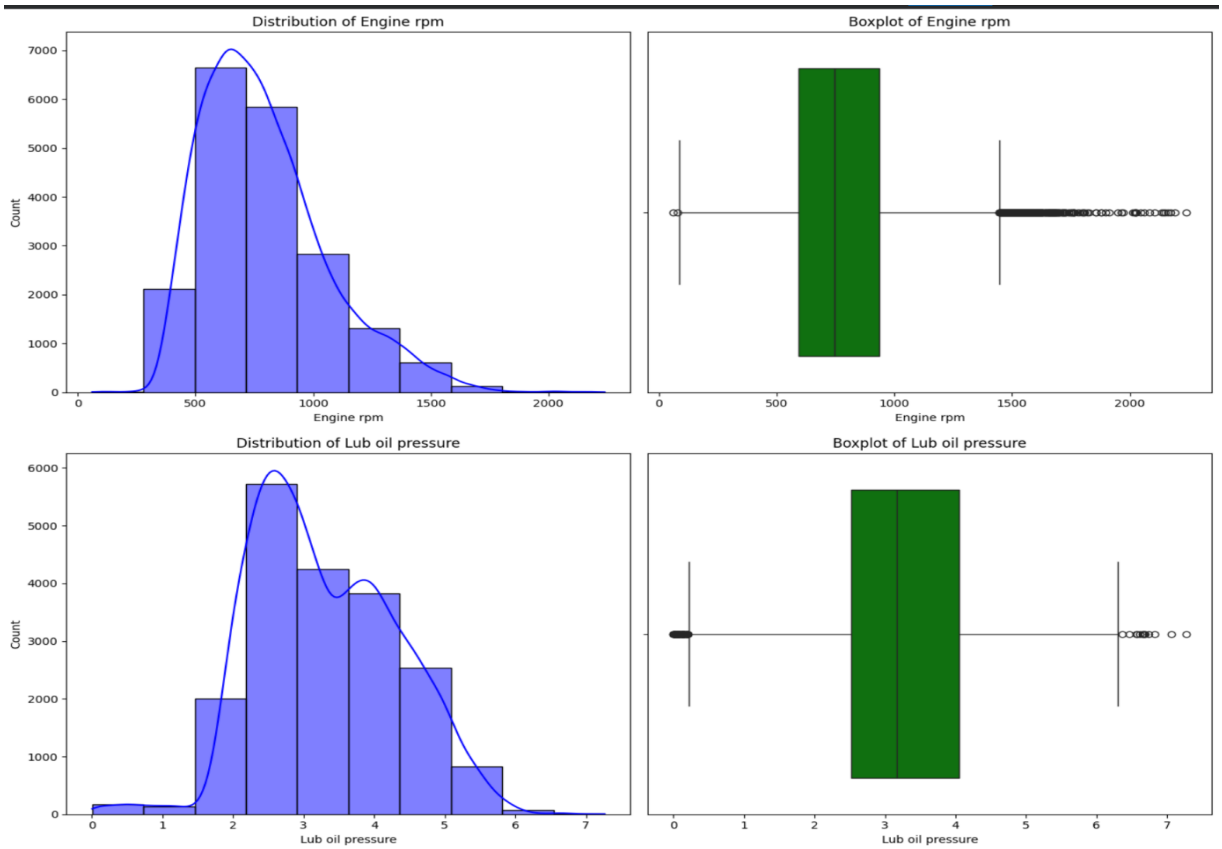

Figure 1 - Boxplot and Distribution of Features

Key Results

- **IQR-Based Outlier Detection**

  The IQR method was applied to each feature, creating binary columns where 1 indicates an outlier. Features like lubricating oil temperature (13.4%) and fuel pressure (5.81%) had the most outliers, while coolant temperature (0.01%) and oil pressure (0.34%) were mostly stable [Figure 2].

  By varying the number of features that must simultaneously be outliers, thresholds were set to control total anomalies. Threshold 1 flagged 23.7% of samples, threshold 2 flagged 2.16%, and stricter thresholds (3–6) flagged almost none. Using threshold 2 ensures that the percentage of detected anomalies falls within the desired 1–5% range, highlighting the most significant abnormal samples.

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Engine rpm outliers: 464
Percentage: 2.375223957000256


Lub oil pressure outliers: 66
Percentage: 0.3378551318146916


Fuel pressure outliers: 1135
Percentage: 5.810084463782954


Coolant pressure outliers: 785
Percentage: 4.018428461735347


lub oil temp outliers: 2617
Percentage: 13.396467878167392


Coolant temp outliers: 2
Percentage: 0.010238034297414897


threshold 1 -> 23.73%
threshold 2 -> 2.16%
threshold 3 -> 0.06%
threshold 4 -> 0.00%
threshold 5 -> 0.00%
threshold 6 -> 0.00%
```

Figure 2

- **Anomaly Detection Using One-Class SVM and PCA**

  To detect anomalies in high-dimensional feature data, a One-Class SVM with an RBF kernel was applied, with gamma set to "scale" and the nu parameter tuned to produce an anomaly rate within the required 1–5% range. Before modeling, the data was scaled, and PCA was applied separately for visualization purposes.

  The 2D PCA plot [Figure 3 - PCA Plot for One-Class SVM] shows most points clustered together, with anomalies detected by SVM clearly separated from normal behavior. After testing multiple values, nu = 0.01 was selected [Table **2** - Summary of One-Class SVM Experiments], flagging approximately 1.01% of samples as anomalies. This ensures that the model focuses on the most significant deviations while avoiding overestimation of outliers.

  Combining PCA and SVM allows both automated detection of unusual patterns and intuitive visualization of anomalous points.

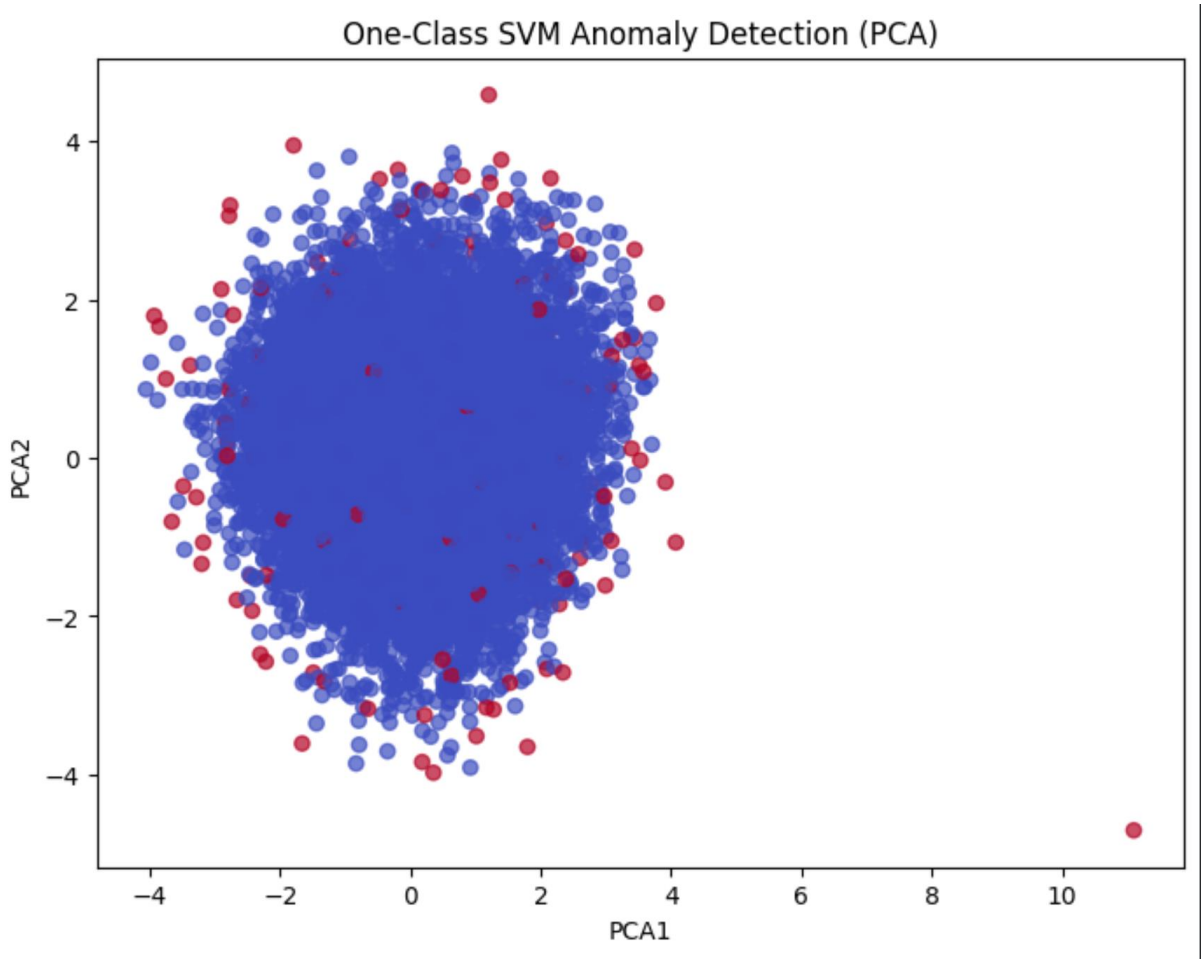

Figure 3 - PCA Plot for One-Class SVM

Nu	Gamma	Outliers	Outlier %
0.005	scale	118	0.604044
0.010	scale	198	1.013565
0.020	scale	400	2.047607
0.030	scale	585	2.994625
0.040	scale	785	4.018428
0.050	scale	976	4.996161

*Table **2** - Summary of One-Class SVM Experiments*

- **Anomaly Detection Using Isolation Forest and PCA**

  Isolation Forest was applied to detect anomalies in the scaled feature space, with PCA used to project the data into two dimensions for visualization. The dataset was standardized using StandardScaler prior to modelling, and PCA was applied once on the same transformed data for visualization purposes.

  The PCA projection [Figure 4 - PCA Plot for Isolation Forest] shows a dense central cluster of observations, with anomalies highlighted by the model. However, there is no clear separation or clustering between normal and anomalous points in the reduced feature space, indicating that anomalies are embedded within the same distribution as normal observations.

  The contamination parameter, which controls the expected proportion of anomalies, was tuned within the 1–5% range. The best-performing setting (contamination = 0.01) identified approximately 1% of the dataset as anomalies [Table 3 - Summary of Isolation Forest Experiments], focusing on the most significant deviations without overestimating outliers.

  Overall, Isolation Forest provides a robust method for anomaly detection, complementing the One-Class SVM results while allowing intuitive visual interpretation through PCA.

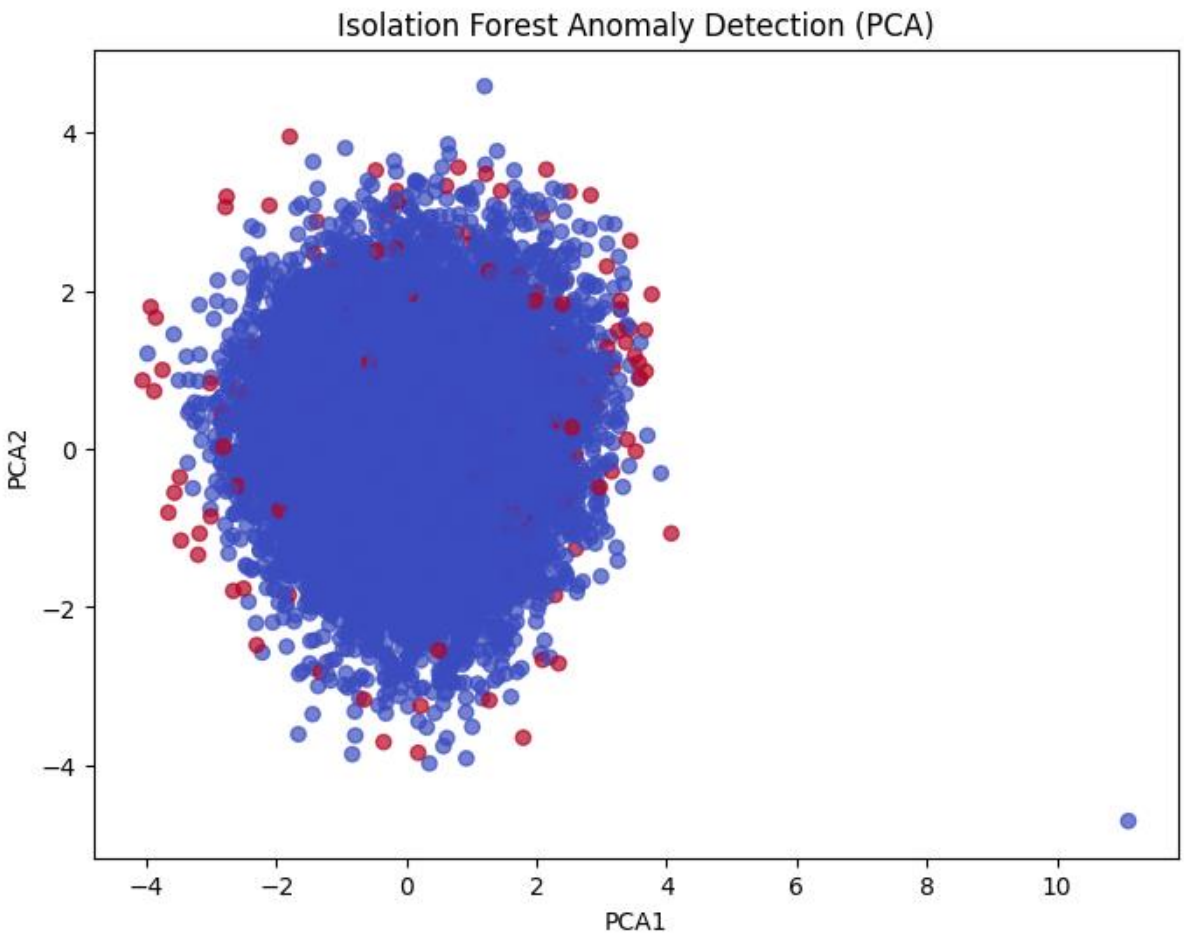

Figure 4 - PCA Plot for Isolation Forest

Contamination	Outliers	Outlier %
0.01	196	1.003327
0.02	391	2.001536
0.03	587	3.004863
0.04	782	4.003071
0.05	977	5.001280
0.06	1173	6.004607

Table 3 - Summary of Isolation Forest Experiments

Comparison of SVM and Isolation Forest with PCA

Both One-Class SVM and Isolation Forest were applied to scaled sensor data, with PCA used once on the same transformed feature space to reduce the data to two dimensions for visualization. The resulting PCA plots show no clear separation between normal and anomalous observations, indicating that anomalies are embedded within the same feature distribution as normal data.

Both models were tuned to flag approximately 1% of samples as anomalies (nu = 0.01 for SVM and contamination = 0.01 for Isolation Forest), keeping detection within the desired 1–5% range. While One-Class SVM tends to define a tighter boundary around the dense region of normal data, Isolation Forest is more effective at identifying isolated extreme observations.

Overall, both methods provide complementary perspectives on anomaly detection, improving the robustness of identifying unusual engine behavior when used together.

Interpretation of the Anomaly Detection Results

Both anomaly detection methods consistently identified a small proportion of samples as abnormal, indicating that most engine operations fall within normal operating ranges.

However, the PCA visualizations show no clear separation or clustering between normal and anomalous observations, suggesting that anomalies are embedded within the same feature distribution rather than forming distinct groups. This indicates that unusual engine conditions arise from combinations of feature deviations rather than single dominant variables.

The agreement between statistical methods (IQR) and machine learning approaches (One-Class SVM and Isolation Forest) suggests that the detected anomalies are meaningful deviations rather than random noise. This consistency improves confidence in the robustness of the results.

Although PCA is useful for visual interpretation, reducing the data to two dimensions results in information loss, meaning that not all anomalies can be visually separated in the projected space.

Recommendations

Based on the results, it is recommended to continuously monitor engine features using automated anomaly detection methods. Both Isolation Forest and One-Class SVM produced consistent results within the desired anomaly range and successfully identified abnormal engine behavior.

For future improvement, incorporating additional sensor features and exploring more advanced nonlinear dimensionality reduction techniques may enhance detection performance. Deploying these methods in a real-time monitoring system would enable early identification of engine issues, reduce maintenance costs, prevent failures, and improve overall operational safety.